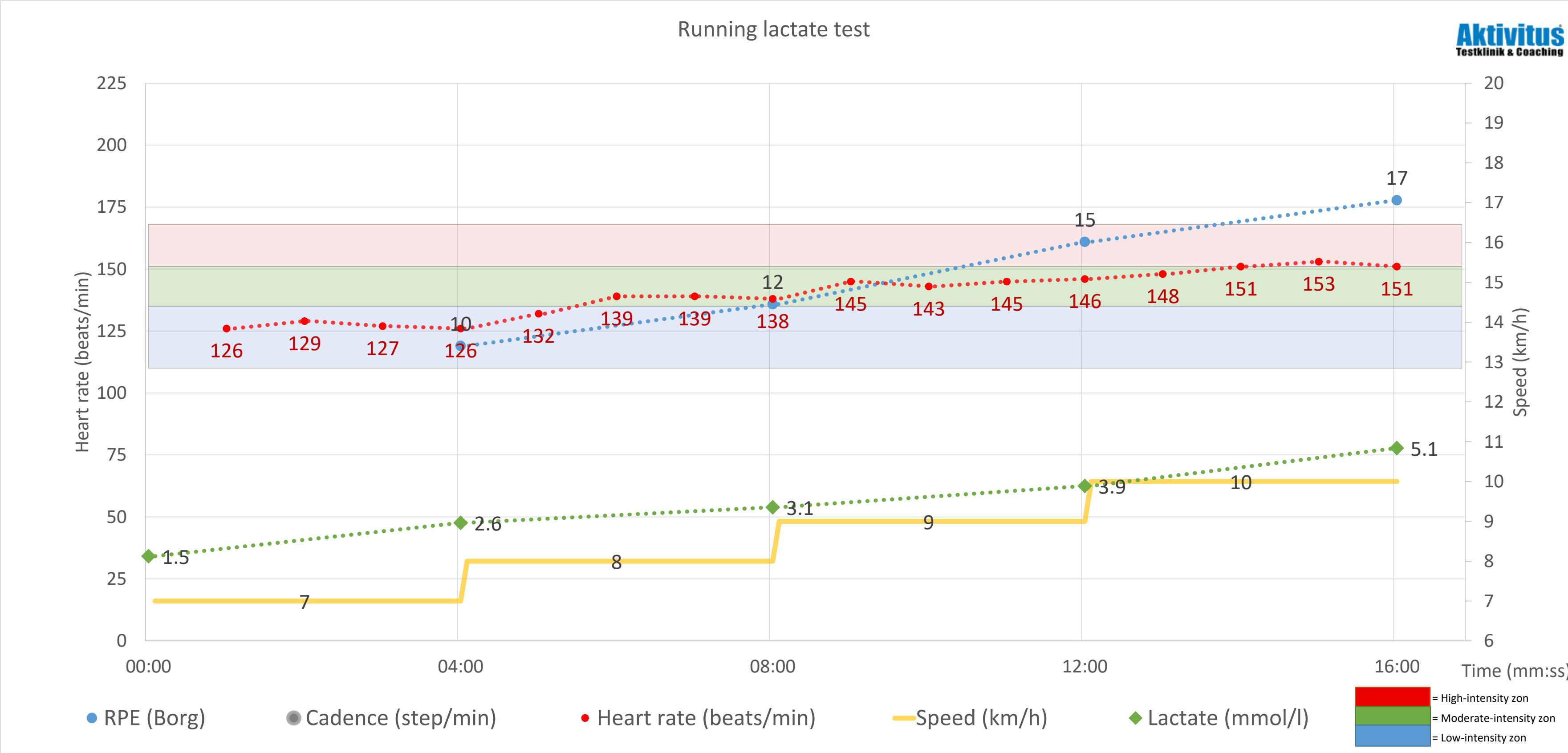
Aktivitus
Testklinik & Coaching

Test report

2026-08-10

Miguel Zavala

Name:	Miguel Zavala			Start speed:	7,0	km/h
Gender:	Male					
Born	1984-08-01			Increase in speed:	1,0	km/h
Age:	42	age		Interval duration:	4:00	min
Weight:	90,0	kg				
Height:	169	cm				
Test administrators:	Axel Mattson					



Anaerob threshold (AT):	
Speed:	9,8 km/h
Pace:	6:05 min/km
Threshold HR:	151 beats/min

Aerob threshold (LT), FatMax:	
Speed:	7,0 km/h
Pace:	8:30 min/km
Threshold HR:	135 beats/min

Max heart rate:	
Estimate:	168 beats/min
Highest HR during test:	163 beats/min

Intensitetszoner:	
High-intensity zon:	151 -168 beats/min
Moderate-intensity zon:	135 -151 beats/min
Low-intensity zon:	110 -135 beats/min

Intensity zones
On the left you will find your mesured Aerobic (LT) and Anaerobic threshold (AT), respectively.

Your individually measured thresholds provide your overall intensity zones in the 3-zone model. Low-, moderate-, and high-intensity heart rate zones. For a more detailed intensity classification, see the next page.

Heart rate, lactate threshold and intensity zones

Lactate thresholds
Aerobic threshold (LT), FatMax
The first physiological threshold is determined where blood lactate increases significantly from baseline level. This change marks the intensity at which fat oxidation begins to decline and thus affects endurance at submaximal exercise intensity. At higher intensities, possible training duration is strongly influenced by the availability of carbohydrates, while fat oxidation in this context represents an unlimited resource. This therefore constitutes an appropriate intensity for long-distance training.

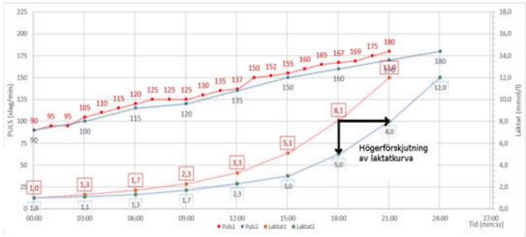
Anaerobic threshold (AT)
The second physiological threshold marks the highest possible intensity at which the body works with sufficient oxygen supply and where lactate production does not exceed what the body can manage, maintaining equilibrium. At AT intensity, the muscles can work for a long time provided that carbohydrates are available as fuel, and your performance in training and competition durations exceeding 30 minutes is largely determined by your AT.

Intensity zones (heart rate and power)
High-intensity training:
What: Interval training consisting of shorter periods of exercise (maximum approximately 8 minutes) above AT. Training in this zone is very to maximally demanding. An effective interval session can consist of 15–40 minutes of effective interval time, excluding recovery periods.
Training effect: Very high training effect on aerobic capacity, but relatively long recovery time.

Moderate-intensity training:
What: Hard training at a steady pace in long intervals or as continuous work for 30–180 minutes depending on intensity. This training is demanding to very demanding, and it is in this zone that you can perform during longer competition and training sessions.
Training effect: The upper part of the moderate-intensity zone provides adaptation and training effect at your race pace. This is very valuable. Training in the lower part of the zone often involves too high a load to train endurance and too low a load to train at race pace. Recovery time is generally shorter than for training in the high-intensity zone.

Low-intensity training:
What: Training at a comfortable pace, slightly demanding. Here you can train for more than 180 minutes. Fatigue results from duration rather than intensity.
Training effect: Good effect on adaptation of muscles, tendons, joints, and skeleton to prolonged work, as well as on general endurance and fat oxidation. Low-intensity training is needed to balance high-intensity training and can be performed as longer distance sessions and shorter recovery sessions.

Example of a precision-controlled interval session from the Monark LC6. 6 × 4 minutes with 2 minutes of rest. The red curve represents heart rate, the green cadence, and the blue power.



Aktivitus Intensity Zones for Those with Membership and Training Programs
For those who have a training program in any form, sessions and intensities are specified according to the scale below. If you want to enter heart-rate zones into, for example, your training watch, use Zone 1 to Zone 5. Z4- and Z4+ together constitute Zone 4. Clarification of each zone can be found in the Aktivitus compendium for workout descriptions that you receive with your program. If you are interested in coaching and a detailed individually tailored training program, see www.aktivitus.se/membership.

The stated running paces for each intensity should be regarded as guideline values. In practical training, pace cannot be maintained as precisely as the zones indicate. As you improve, your running speeds will increase. If you are running at a pace that feels easier than usual and your heart rate is low, it is time to increase your pace zones. When running on hilly routes or in terrain, primarily use heart rate and perceived exertion.

Z8 MAX	Not applicable	MAX Acceleration/Sprint		
Z7 PR	Not applicable	5,05	and above	min/km*
Z6 TO	Not applicable	5,24	5,05	min/km*
Z5 KI	156 -168 beats/min	5,45	5,24	min/km
Z4+ T+	151 -156 beats/min	6,09	5,45	min/km
Z4- T-	146 -151 beats/min	6,47	6,09	min/km
Z3 D	135 -146 beats/min	8,34	6,47	min/km
Z2 LSD	110 -135 beats/min	11,26	8,34	min/km
Z1 R	Up to -110 beats/min	Up to	11,26	min/km

Maximal oxygen uptake / VO2max

Protocol and values

Protocol:

Speed	8	km/h
Incline increase:	2	%/min
Total time:	05:30	mm:ss
Max incline:	8	%
Test value:	**44,6**	ml O_2/kg/min
Oxygen uptake:	**4016**	ml O_2/min

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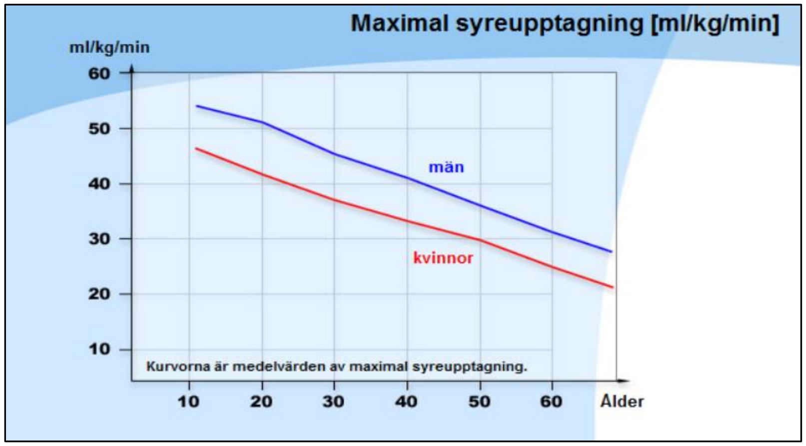

Test of maximal oxygen uptake (VO_2max)

During a VO_2max test, the maximum amount of oxygen that the body can utilize during aerobic metabolism is measured. At VO_2max, anaerobic energy processes are also fully active, and therefore the power output or running speed reached at VO_2max is higher than what is theoretically required to correspond solely to aerobic metabolism. Maximal oxygen uptake capacity is often expressed as absolute capacity (L/min), but since most endurance sports involve supporting one's own body weight, a relative value is commonly used, the so-called test value (ml/kg·min^{-1}). Success in endurance sports generally requires a high maximal oxygen uptake capacity, and VO_2max for women and men competing at elite level is most often within 60–75 and 65–85 ml/kg·min^{-1}, respectively. A lower value can be compensated for by higher efficiency and a higher utilization rate (anaerobic lactate threshold). In physiology, oxygen uptake capacity is divided into central and local capacity.

Central capacity consists of the heart's pumping ability and the blood's ability to transport oxygen. Central capacity largely determines VO_2max.

Local capacity consists of how oxygen is taken up and utilized in energy metabolism in the working musculature. The muscles generally have an excess capacity to utilize oxygen and do not limit VO_2max during whole-body exercise that engages a large muscle mass. However, the muscles are limited by decreasing energy availability and the development of fatigue. A high local capacity therefore results in good endurance.

Central capacity can be trained using any method that places a high blood flow load on the heart. Local capacity must be trained in a discipline-specific manner, as the training effect is linked to how and which muscles are working.

Average values for VO_2max testing at Aktivitus (all ages):

Women: 48 ml/kg

Men: 54 ml/kg